

LAB Assignment-5 Answers

1. Display fname ascending:

```
SELECT first_name AS Name FROM emp ORDER BY first_name ASC;
```

2. Display fname descending:

```
SELECT first_name AS Name FROM emp ORDER BY first_name DESC;
```

3. Display hire date ascending:

```
SELECT hire_date FROM emp ORDER BY hire_date ASC;
```

4. Employees whose fname starts with J or M:

```
SELECT * FROM emp WHERE first_name LIKE 'J%' OR first_name LIKE 'M%' ORDER BY last_name;
```

5. Salary stats:

```
SELECT MAX(sal) AS Max, MIN(sal) AS Min, AVG(sal) AS Avg, SUM(sal) AS Sum FROM emp;
```

6. Salary stats per job:

```
SELECT job_id, MAX(sal), MIN(sal), AVG(sal), SUM(sal) FROM emp GROUP BY job_id;
```

7. Number of people under each job:

```
SELECT job_id, COUNT(*) FROM emp GROUP BY job_id;
```

8. Number of managers:

```
SELECT COUNT(DISTINCT manager_id) FROM emp;
```

9. Difference between highest and lowest salary:

```
SELECT MAX(sal) - MIN(sal) AS Diff FROM emp;
```

10. Max & avg salary of engineers:

```
SELECT MAX(sal), AVG(sal) FROM emp WHERE job_id LIKE '%ENG%';
```

11. First and last fname alphabetically:

```
SELECT MIN(first_name), MAX(first_name) FROM emp;
```

12. First and last hire date:

```
SELECT MIN(hire_date) AS First_Hire_Date, MAX(hire_date) AS Last_Hire_Date FROM emp;
```

13. Max & avg salary of clerks:

```
SELECT MAX(sal), AVG(sal) FROM emp WHERE job_id LIKE '%CLERK%';
```

14. Lowest salary per dept:

```
SELECT dept_id, MIN(sal) AS Min_Salary FROM emp GROUP BY dept_id;
```

15. Lowest salary >3000 per dept:

```
SELECT dept_id, MIN(sal) AS Min_Salary FROM emp GROUP BY dept_id HAVING MIN(sal)>3000  
ORDER BY Min_Salary DESC;
```

16. Display formatted salary statement:

```
SELECT first_name || ' whose designation is ' || job_id || ' gets ' || sal || ' but wants to earn ' || sal*3  
FROM emp;
```

17. Display all details in single line:

```
SELECT emp_id || ', ' || first_name || ', ' || last_name || ', ' || hire_date || ', ' || job_id || ', ' || sal || ', ' ||  
dept_id FROM emp;
```

18. Today's date:

```
SELECT SYSDATE AS Today FROM dual;
```

19. Employee id, day, year hired:

```
SELECT emp_id, TO_CHAR(hire_date,'Day'), TO_CHAR(hire_date,'YYYY') FROM emp;
```

20. Hire date 'dd-month-yy':

```
SELECT first_name, TO_CHAR(hire_date,'dd-month-yy') FROM emp;
```

21. Months employees were hired:

```
SELECT emp_id, TO_CHAR(hire_date,'Month') FROM emp;
```

22. Hire date 'dd-mon-yyyy':

```
SELECT first_name, emp_id, TO_CHAR(hire_date,'dd-mon-yyyy') FROM emp;
```

23. Hire date 'month-dd-yyyy':

```
SELECT first_name, emp_id, TO_CHAR(hire_date,'month-dd-yyyy') FROM emp;
```

24. Display system year in words:

```
SELECT TO_CHAR(SYSDATE,'Year') FROM dual;
```

25. Dates 15 days before and after:

```
SELECT SYSDATE+15 AS After_15, SYSDATE-15 AS Before_15 FROM dual;
```

26. Apply domain and integrity constraints:

```
CREATE TABLE emp(...);
```

27. Drop domain constraint:

```
ALTER TABLE emp MODIFY sal NUMBER;
```

28. Drop primary key:

```
ALTER TABLE emp DROP PRIMARY KEY;
```

29. Drop table:

```
DROP TABLE emp;
```